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Research paper

Scan-to-BIM-to-Sim: Automated reconstruction of digital and simulation models from point clouds with applications on bridges

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A R T I C L E I N F O

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A B S T R A C T

The automation of 3D geometric model reconstruction from point clouds is essential for efficient management of critical infrastructure assets like bridges, significantly streamlining and enhancing inspection and structural analysis tasks. However, existing automated frameworks frequently encounter challenges due to substantial computational demands and the difficulties when applied to defective point clouds, which arise from adverse environmental conditions, measurement errors, and limitations of surveying equipment. Major limitations also exist in achieving real-time data exchange and mapping between Building Information Modelling (BIM) models and simulation (Sim) models. To address this gap, this paper proposes a comprehensive Scan-to-BIM-to-Sim framework that efficiently reconstructs bridge BIM models from imperfect point clouds and supports bidirectional mapping between BIM and numerical simulations. The method proposes an improved edge detection for contour extraction from imperfect point cloud and parametric modelling for the direct generation of 3D models within BIM software. Additionally, it automates the exchange between BIM models and simulation software, facilitating bidirectional operation for real-time analysis and visualisation. The framework, validated using the Arial Aqueduct Bridge case, reduces modelling time and computational demands, thereby streamlining construction simulations with improved accuracy and cost efficiency. The dataset collected for model generation and validation is openly available at https://doi.org/10.17632/znxxsgn2ky.1.

1. Introduction

Digital Twin technology which has been increasingly utilised in bridge engineering, has demonstrated significant potential to automate and streamline engineering tasks throughout the asset lifecycle [1–4]. To reconstruct a digital model, scanned bridge data, including images and point clouds, must be transformed into visualised 3D models. These models can then be updated to facilitate bridge health monitoring and management, thereby reducing maintenance and operational costs [5–7]. Converting bridge point cloud data into semantically enriched BIM models enhances bridge asset management by providing accurate 3D representations and visualisations of structural bridge components. Additionally, digital models can be imported into finite element (FE) simulation software to allow for more automated structural analysis. This process has been proven to be effective in building engineering and tunnel engineering [1,8–11]. These FE simulations can subsequently be utilised to assess load-bearing capacity and swiftly perform damage characterisation based on predefined criteria, in support of non-destructive testing.

However, current research lacks a unified, comprehensive framework to guide the process of converting scanned data into semantically enriched BIM models, followed by numerical analysis [12], and ultimately mapping the results back to the BIM model [13]. Furthermore, BIM and numerical models typically utilise static data, whereas bridge maintenance and management require dynamic data support. As a result, there is a need to effectively integrate monitoring data into BIM and numerical simulations, extracting relevant structural information based on analysis objectives, and performing efficient data transformation and integration to enable real-time structural health monitoring and assessment. Additionally, current Scan-to-BIM technology primarily relies on high-quality data for reverse modelling, yet in practical engineering applications, it is often necessary to deal with low-quality or defective data due to the quality of scans to point clouds. In these cases, existing methods tend to be less accurate and hence of reduced effectiveness.

This paper introduces two main contributions: firstly, a new accurate and effective method specifically designed for defective point cloud data to generate BIM Models is developed. The method provides an

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alternative for model reconstruction in the absence of high-quality point clouds. Secondly, an automated bridge reconstruction framework is proposed. This encompasses the whole process from acquiring bridge point cloud data, through generating 3D geometric models, to producing finite element models (Scan-BIM-Sim). The framework for automated reconstruction and analysis is illustrated in Fig. 1.

This framework is divided into two stages. The initial stage, termed the Scan-to-BIM conversion, involves transforming scan data (point cloud) into a parametric BIM model that incorporates updatable and semantic elements. The subsequent stage, designated as BIM-to-Sim interaction, facilitates the exchange between the BIM model and the simulation model. In this interaction, the BIM model transmits geometric and semantic data to the simulation model, which then processes and returns the results for visualisation and prospective retrofitting solutions. Comparisons between modelled deformations and the original scanned data are performed to validate structural deformations. Diverging from previous frameworks, our framework facilitates data circularity, model automation, and upgradability, thereby enhancing the framework's resilience.

2. Related work

Methods for constructing geometric models from point cloud data include manual and automated modelling. Manual modelling involves measuring component dimensions and manually creating and assembling the model in BIM software. In contrast, automated modelling uses programming to extract shape information from bridge cross-sections by importing point cloud data and generating geometric models through scripts that interact with BIM software. Given the labour-intensive and expensive characteristics of manual modelling, recent research has increasingly focused on developing automated modelling techniques [14].

2.1. 3D geometric model reconstruction

The reconstruction methods for bridge 3D geometric models can be classified into geometry-based and learning-based approaches. Since bridges typically exhibit regular polyhedral geometry, their shape can be described using geometric features such as planes, edges, curves, and corners [15]. The Random Sample Consensus (RANSAC) method [16–19] is commonly employed to determine the equations of planes or edges within point clouds, while Non-Uniform Rational B-Splines (NURBS) [20–22] are utilised to fit the complex contours of bridges. Point histograms [23] and curvature clustering techniques [24,25] are applied to extract feature points from the bridge outline. Alternatively, optimisation algorithms can directly fit the overall contour of the bridge cross-section [26–28], with the objective of minimising the distance between each point and the fitted contour. In addition, edge detection techniques [29–31] in computer vision can be adapted for point clouds by projecting the data onto a 2D image, where edge detection operators are used to extract edges [32]. The characteristic of edge pixels having significantly higher grey values than surrounding pixels allows edge detection operators, such as the Sobel [33], Canny [34], and Roberts [35] operators, to convolve the grey values of each pixel, thereby detecting variations in grey levels and extracting the image outline. However, the projection to an image can result in the loss of point cloud information, which may compromise the accuracy of the results.

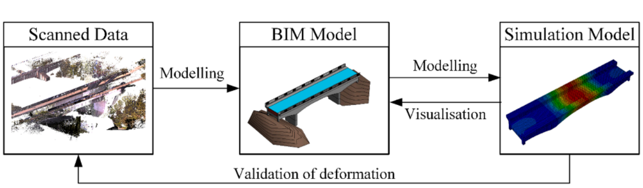


Fig. 1. Resilience-Enhancing Scan-to-BIM-to-Sim Workflows.

Furthermore, the aforementioned methods are sensitive to noise and outliers in point clouds, meaning that satisfactory performance is often not achievable with low-quality point clouds.

Learning-based methods primarily refer to machine learning or deep learning techniques, where Neural Networks (NN) are employed to learn the features of points to be recognised [36,37], and generally require large datasets. Classical point cloud neural networks, such as PointNet++ [38,39], Dynamic Graph CNN (DGCNN) [40], PointCNN [41], and PointConv [42], can perform point cloud classification by extracting high-dimensional features for each point. Specifically, in tasks involving parameter model construction, these networks can be used to distinguish the edge contours and corner points of point clouds. However, these methods need to be based on the models trained with large datasets that are characterised complex and computationally intensive training process. Although NN-based models offer a promising avenue, their widespread application remains limited due to the high cost and technical challenges of obtaining bridge point cloud data, as well as the lack of publicly available datasets and limited data sharing by asset owners.

2.2. Model generation

Currently, the most widely applied method of automatic bridge reconstruction in BIM software is to divide the bridge into its components, reconstruct them individually and assemble into a bridge model. The bridge deck is constructed by slicing it along the centreline according to the given interval, and then placing the instances of the defined cross-section class at the locations of the cross-section slices [21,27].

Overall, the principal challenges in bridge reconstruction stem from the sensitivity to the quality of the point cloud and the necessity to balance the level of detail in bridge modelling with the computational burden. If the bridge 3D model is to be generated with as much detail as possible, more cross-section point clouds and constraints need to be considered, which results in higher computation time. Therefore, there is a need to develop a stable and reliable method that can reduce the computational effort as much as possible, cope with most of the engineering point cloud quality problems (missing data, edge effects, noise, etc.), and extract the bridge parameters effectively and accurately.

Many BIM commercial software have embedded programming tools. For example, Dynamo [43], the visual programming software that comes with Autodesk Revit software [44], supports users to model through programming [45,46]. By developing scripts, it is possible to control the geometric parameters of the bridge by creating and editing nodes to achieve interactivity in the model [47]. The approach can not only intuitively obtain the changes of the parameters of the bridge at each location by defining the parameters but also supports the subsequent adjustment of the model to achieve the variability of the model. Hence this method is adopted in this study to achieve the reconstruction of the bridge.

2.3. Integrated design-through-analysis workflows

Even though the interoperability between BIM and any simulation software, in theory, should be realised through common data formats such as Industry Foundation Class (IFC), the current numerical software implementations lag behind. To bridge current technology gap, interoperability between the BIM with numerical analysis software is achieved through other specific file formats. For instance, the Standard ACIS Text (SAT) file format, which is used to represent and exchange 3D geometry data among various applications, can achieve interoperability with Revit [48]. However, this approach only facilitates the import of 3D solid models, limiting the transfer of additional information such as rebars or materials information. Subsequent operations such as the definition of material parameters, boundary conditions, etc. need to be defined by user defined script for automatic operation or manually

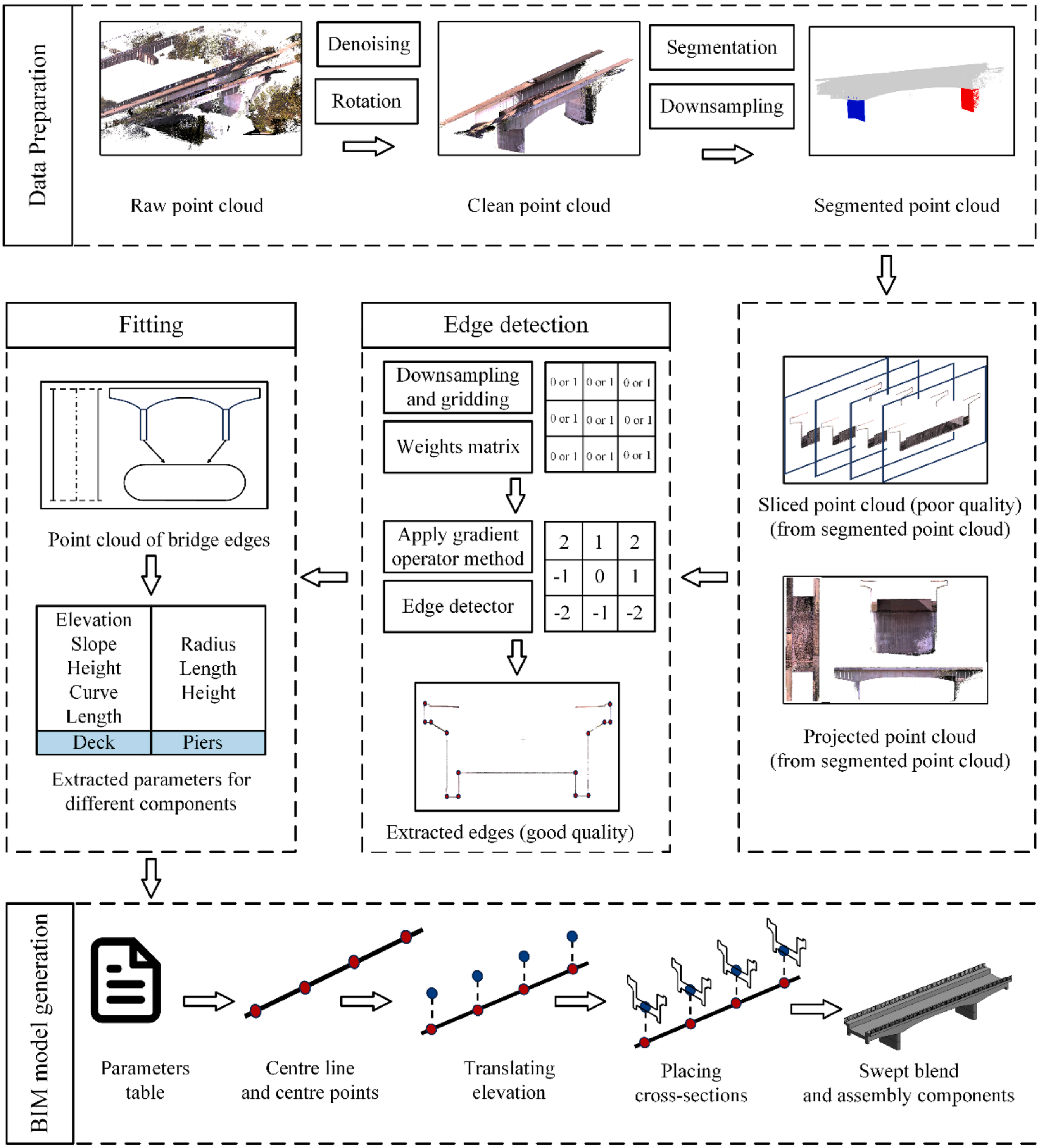


Fig. 2. Framework of Scan to BIM.

defined. Since Finite Element Modelling (FEM) software such as ABAQUS [49] supports the execution of user commands via Python code, a common method to realise BIM-FEM interactivity is to develop scripts for data exchange, and to realise the mapping between FEM results and BIM through code blocks.

FE software generally offer a Python interface that allows for extensive customization and automation of tasks. In this interface, every operation within the simulation software is represented as interactions between various classes and functions [50]. Each step, whether it involves defining geometry, applying boundary conditions, or post-processing results, is implemented through object-oriented code. Typically for simple geometries, executing simulation through a Python script will simplify the operation and reduce the operation time. However, for models with complex geometries and tasks that require further division into sets and partitions, using scripts will increase the time and operation steps.

Several studies have been developed to automate the BIM-Sim interaction frameworks in tunnelling engineering [50–52], through which it is possible to automate the creation of 3D solid models of tunnels in Dynamo for Revit, and to transfer them between BIM and FEM

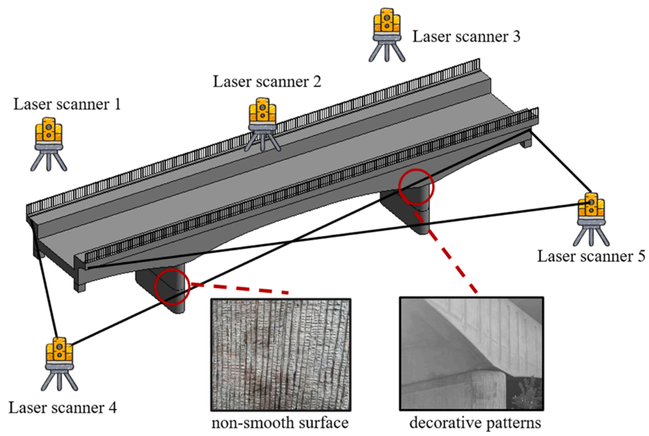


Fig. 3. Layout of the laser scanners and detail of the bridge surface.

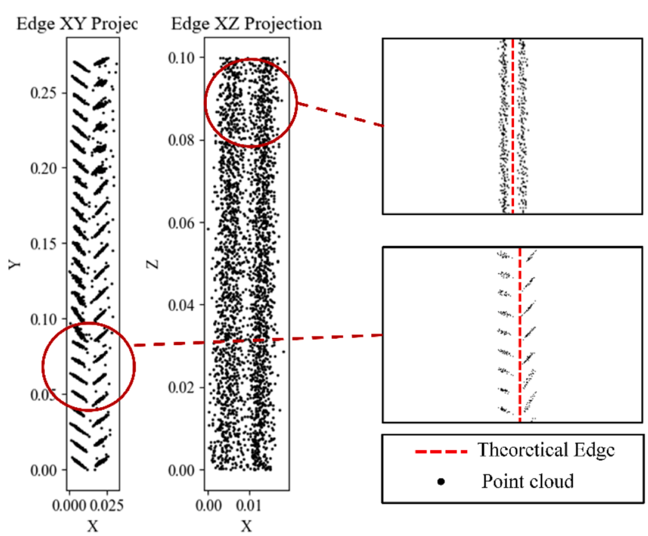


Fig. 4. Layered point cloud of bridge contours.

software [51,53]. These frameworks enable interaction between BIM and FEM, while automating tasks, such as material property assignment, construction simulation and high-quality mesh generation [54]. However, the geometry of tunnels is relatively simple, and the analysis steps and boundary conditions are easier to define using code. For large and complex structures such as bridges, managing intricate geometries presents significant challenges. Additionally, numerical models simulate prestressed and conventional reinforcement rebars in distinct indirect manners, precluding direct importation from BIM systems. Consequently, the Scan-to-BIM process for bridges predominantly depends on manual interventions in the user interface, complicating the automation of data transfer and integration.

2.4. Problem statement and objectives

The literature review highlights that current methods for 3D geometrical model reconstruction and analysis are complex, costly, and computationally intensive. There is a notable absence of a comprehensive workflow that seamlessly integrates 3D geometric model reconstruction, BIM model visualisation, and numerical analysis to facilitate flexible data exchange. Furthermore, these methods depend on high-quality point clouds, which must have minimal noise and clearly defined boundaries and contours. This dependency renders them unsuitable for bridges with intricate patterns or ornamental details, as these features typically result in rough and imprecise edges in scanned

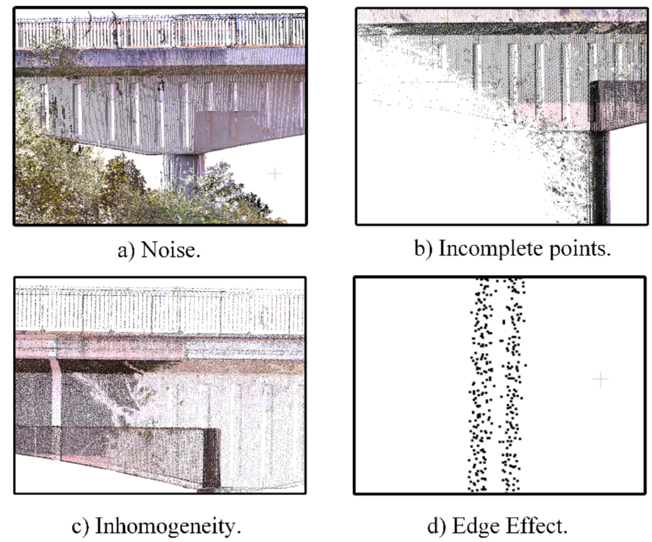


Fig. 5. Quality defects of point clouds.

data. Additionally, numerical modelling tools lack effective strategies to reduce manual input from digital models, automate data importation, and precisely define models, further limiting their applicability in complex scenarios.

To bridge these gaps, this paper introduces a novel Scan-to-BIM-to-Sim framework for bridges. This framework optimises the complexity of implementation while maintaining adequate accuracy. Utilising a defective point cloud, a modified edge detection operator is employed to segregate layered edges and derive bridge parameters by pinpointing corner points. Automated scripts then facilitate the seamless transfer of these parameters into a solid BIM model. Additionally, the automation extends to defining materials, boundary conditions, and forces through specifically created classes for each component, streamlining the transition from BIM solid models to real-time finite element (FE) analysis. Moreover, our framework allows for bidirectional interaction between BIM and numerical simulations, enabling real-time analysis and visualisation. This enhancement is vital for thorough bridge condition assessments and diagnostics. While the framework was validated in an aqueduct case study, the methodology is generalisable and can be applied to any bridge type, directly utilising raw point clouds. This approach reduces dependence on extensive point cloud processing and expands the framework's applicability across a broader range of scenarios.

3. Methodology

The first stage of the framework (Scan-to-BIM) can be further divided into four steps (illustrated in Fig. 2): **Step1**: data preparation, **Step2**: edge detection on point cloud data, **Step3**: parameters extraction and **Step4**: model generation.

The asset is initially parameterised using a set of geometric

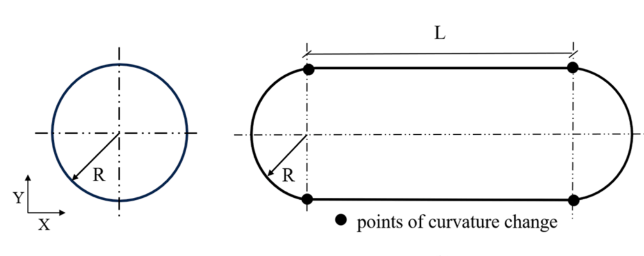


Fig. 6. Parameter of circular pier (a) and combined shape pier (b).

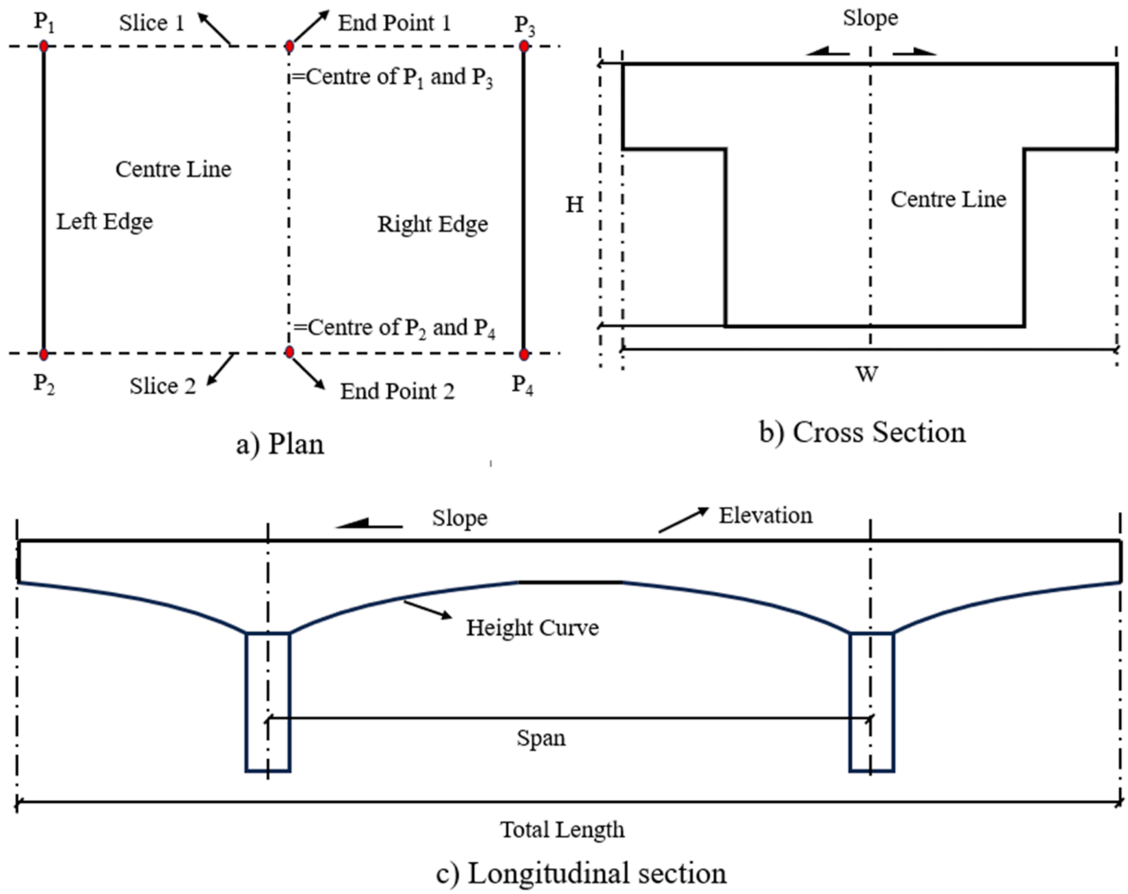


Fig. 7. Information extracted from different sections.

attributes, such as height, length, width, slope, and elevation, which uniquely define the bridge profile's shape. Adjusting the values of the parameters allows for a straightforward control of the cross-section shape and size. The original point cloud is first denoised and segmented, then sliced and projected. After performing edge extraction on the projected and sliced point cloud, the parameters are extracted from the extracted edges by the proposed algorithm. The cross-section parameters are extracted from the sliced point cloud, the longitudinal section and other parameters are extracted from the projected point cloud. Lastly, the BIM model is generated according to the customised scripts.

3.1. Collection of point cloud data

Commonly used methods for obtaining point clouds are Structure From Motion (SFM), which uses a unmanned aerial vehicles (UAV) mounted camera to capture images that are converted to point clouds [55], and Terrestrial Laser Scanning (TLS) [56,57], which uses a laser scanner placed on the ground to capture the laser light reflected from an object to obtain point cloud information. TLS scanning can accurately generate point clouds in a shorter time, and contains not only coordinate information, but also information such as colour, intensity, and normal vector, which makes it widely used in engineering surveys [58,59]. In this paper, the point cloud of bridge is acquired by the TLS method.

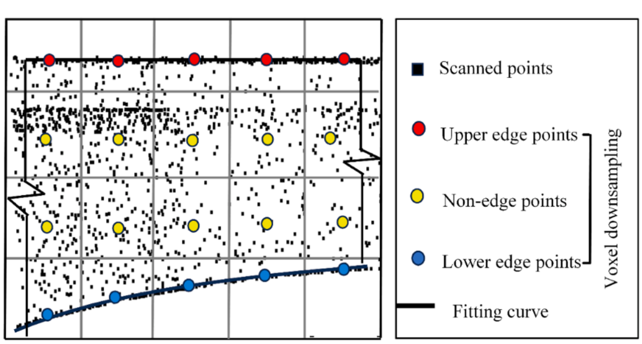


Fig. 8. Grid-based approach to extract the edges.

Generating a point cloud requires multiple scans from different angles to ensure good coverage over the whole feature as line of sight is compromised to parts of the bridge for any single location. Data for the study are collected using a total station with terrestrial laser scanning capabilities (Trimble SX10), which has the advantage of being able to position the location and orientation of the scanner accurately using the standard surveying functions, ensuring that the collected points in the cloud are in a common reference system. To obtain points from all sides of the feature, five different scanning locations are used (two either side of the bridge and one on top) which are located by fore sighting onto the next survey point to determine its location (shown in Fig. 3), before moving the instrument to setup over the new point and back sighting onto the previous point to set the orientation. Such an approach is more accurate than using GNSS derived coordinates and should ensure systematic offsets between scans are <1 cm and the instrument expected accuracy of scan points is 10mm+10 ppm [60]. Scans are collected using the standard density which takes points every 0.5 milliradian, equivalent to 5 mm spacing at 10 m. Overlap between the different scan locations acted as a quality control measure.

Although it is theoretically possible to get an accurate point cloud through the TLS method, in the actual scanning operation, for safety and practical reasons, the terrestrial laser scanner can only be placed on both sides longitudinally on the sides of the bridge. When the laser scans the

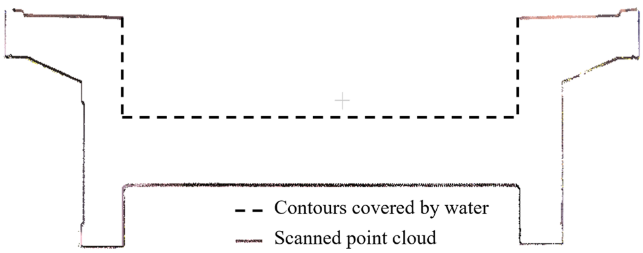


Fig. 9. Cross-section slice of the bridge.

object from two sides (as shown in Fig. 3, scanners 4 and 5), system errors can occur due to the movement of the equipment and the significant differences in the incident angles [61]. These errors may result in layering and roughness in the contour of the point cloud [62]. Additionally, bridges may feature non-smooth surfaces and decorative patterns or carvings for aesthetic purposes or due to construction techniques. Such complex uneven surfaces can cause point cloud overlap when reflecting lasers emitted from different angles [63]. Finally, point clouds obtained using the TLS method are susceptible to weather and lighting conditions, which can degrade their quality [64]. These three types of system errors are commonly found in scanning practices and can accumulate, resulting in a defective point cloud (i.e., thick edges), as shown in Fig. 4.

Therefore, even if the TLS method is used to obtain the point cloud, there is still a probability of acquiring a low-quality point cloud. This paper describes how BIM-based reconstruction is performed using low-quality point clouds in Section 3.3.

3.2. Point cloud data pre-processing

Ideally, the point cloud should be uniform, smooth, and complete. However, in engineering practice, bridges are often situated in complex environments with obstructions such as vegetation and other obstacles, leading to inevitable point cloud data missing that significantly impact quality. Thus, the initial data obtained from laser scanners generally has many defects, such as noise, incomplete points, edge effect, redundancy, inhomogeneity, etc., as shown in Fig. 5. Therefore, it is necessary to pre-process the original point cloud data, such as alignment, denoising, segmentation and transformation operations. To enhance the automation of the framework, the Statistical Outlier Removal (SOR) method is used for point cloud denoising, the Cloth Simulation Filter (CSF) method is applied for ground point removal, and the k-means clustering method is employed for point cloud segmentation in this paper.

To prevent the point cloud coordinates from being too large and causing bias in the subsequent fitting process, the point cloud needs to be translated and rotated to make the ends of the bridge coincide with the x-axis. To rotate the point cloud, a rotation angle θ is required. Principal Component Analysis (PCA) [65] is used to identify the bridge's main direction and compute its angle relative to the coordinate axis, which is then applied as the rotation angle. The point cloud is subsequently rotated accordingly.

3.3. Extraction of bridge parameters

3.3.1. Piers parameters

The shape of bridge pier is relatively simple; hence it can be easily parametrised (see Fig. 6). The formula for extracting parameters from the x, y, and z coordinates for a circle is as follows:

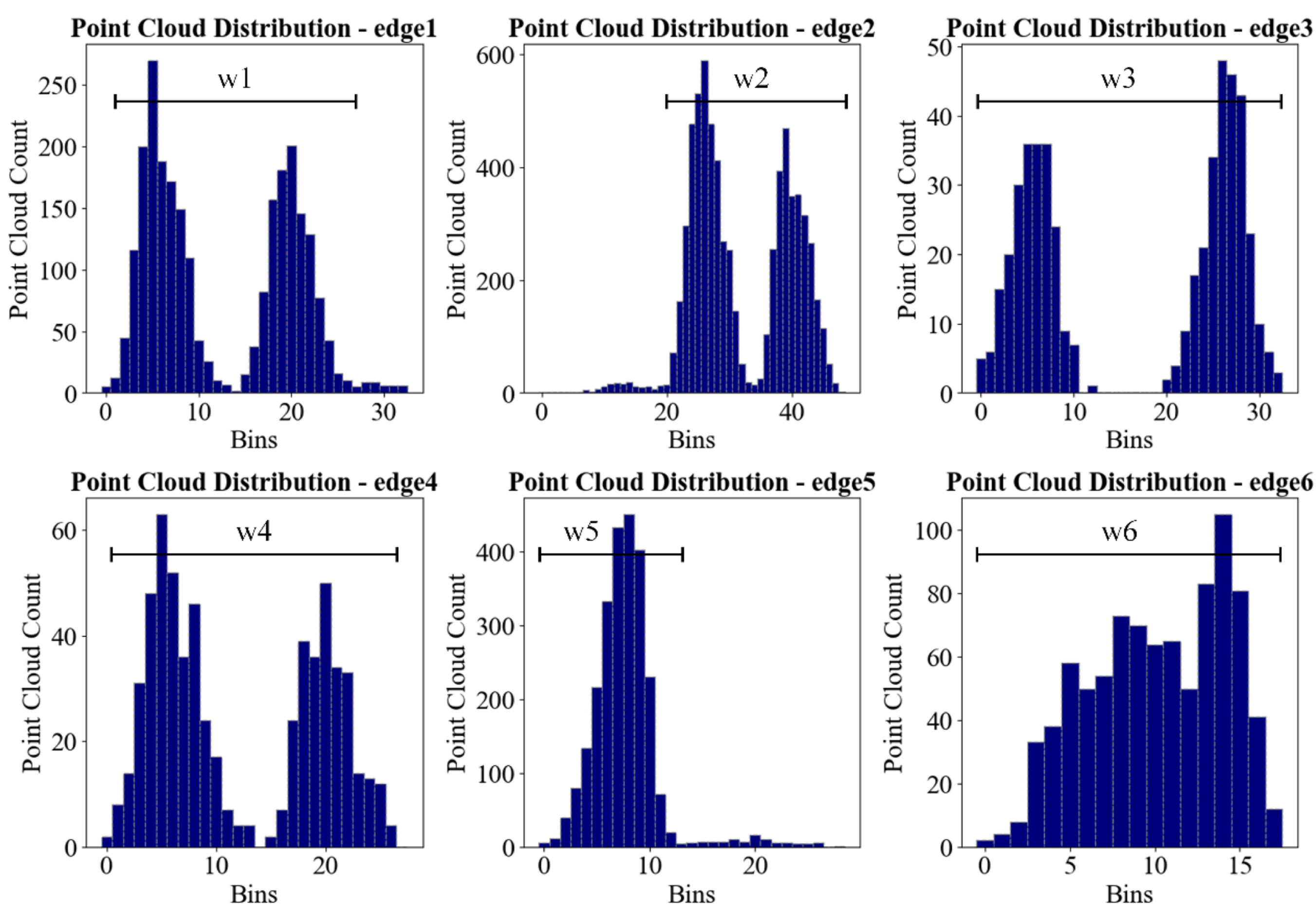


Fig. 10. Histogram of 1 mm size bin of each peak.

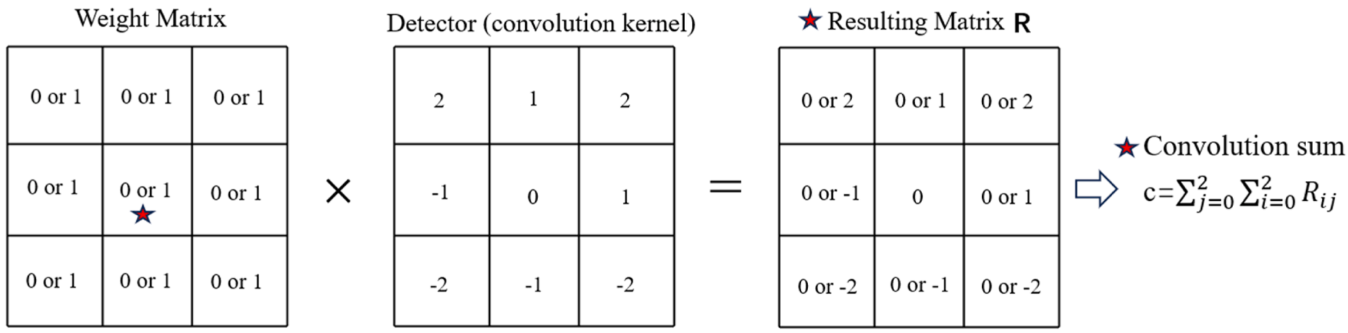


Fig. 11. Grid weight and detector of the edge detection method.

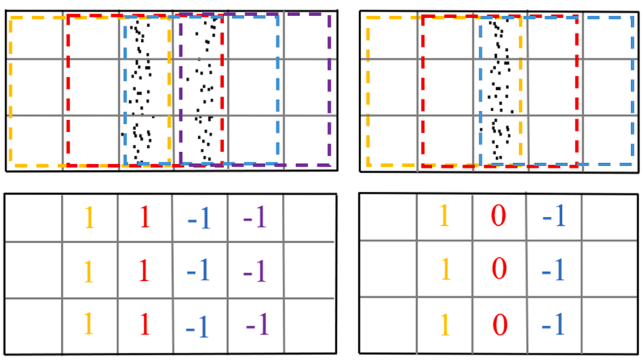


Fig. 12. Possible results for convolution.

$$\begin{cases} R = \dfrac{y_{\max} - y_{\min}}{2} \\ H = \mathrm{Zmax} - \mathrm{Zmin} \end{cases} \tag{1}$$

In Equations (1): R represents the radius, H denotes the height. The cross-section of the combined shape pier is considered to consist of two circular arcs and two straight lines. It is thus segmented into four parts distinguished by points of curvature change for individual fitting. The length of the straight edge L is determined by the distance between two points of curvature change, and R is obtained through fitting.

3.3.2. *Plane curve parameters*

The deck is characterised by a combination of straight lines and curved segments. The cross-sectional outline is defined as a series of intersecting straight lines, while the longitudinal profile of the bridge is defined by a straight line (representing the longitudinal slope of the bridge deck) and a quadratic curve (representing the variation in bridge height).To define the location and geometry of the bridge, the three views of the bridge, i.e. the plan, cross section and longitudinal section of the bridge, can be used (see Fig. 7). The plan is used to determine the coordinates of the bridge centreline, the longitudinal section is used to determine the elevation of the bridge centreline and the longitudinal slope gradient, and the cross section is used to determine the size of the bridge components and the cross-section information parameters.

The extraction of information mainly relies on extracting the edges, dividing the contours and fitting them separately. In the case of a cantilever bridge, for example, the centreline is determined as the line joining the centres of the endpoints of the left and right boundaries, as shown in Fig. 7a. Since the alignment of the bridge may be curved, the bridge is extracted in segments at regular intervals, so that each segment can be regarded as a straight line. The parameters of the cross section are slightly more complex, and the method for extracting the parameters will be elaborated in Section 3.3.3. The longitudinal slope gradient and height curves of the longitudinal section are obtained by fitting the upper and lower boundaries, respectively. The span and length of the bridge are obtained from the histogram of the longitudinal section as the distance between two identical peaks of the histogram.

Due to the disorder and rotational invariance of the point cloud, it is necessary to find a suitable method to segment the boundary point cloud from the point cloud projection. A grid-based approach is used to extract the boundary point cloud.

The bounding box of the point cloud is divided into a grid and numbered (shown in Fig. 8). The points in the point cloud are then allocated to the corresponding grid cells. In this way, each grid cell forms a set containing the points whose coordinates fall within the grid's boundaries. Subsequently, the points within each grid cell are replaced by their centroid. The centroid of the point cloud $P_{centroid}$ ($x_{centroid}$, $y_{centroid}$, $z_{centroid}$) of the grid is calculated according to Eq. (2) [66]:

$$\begin{aligned} x_{centroid} &= \frac{\sum_{i=1}^{m} x_i}{m} \\ y_{centroid} &= \frac{\sum_{i=1}^{m} x_i}{m} \\ z_{centroid} &= \frac{\sum_{i=1}^{m} z_i}{m} \end{aligned} \tag{2}$$

Where x, y, and z are coordinates and m represents the number of points in the grid. The centroid well represents the 3D coordinate information of all the points within the grid, and it can reduce the amount of computation significantly, subsequent operations are based on the centroid in the grid. This method is similar to voxel down sampling, but the key difference lies in associating the centroid coordinates with the grid cell identifiers, which facilitates filtering and classification based on the grid numbering.

As shown in Fig. 8, the point clouds corresponding to the first and last rows of the grid represent the upper and lower edges, respectively. Since the point clouds of bridges obtained from the laser scanner are usually in the millions, the point clouds need to be down sampled first to reduce redundancy and computation time.

Algorithm 1
Extract new edges based on the convolution result.

Input: Convolution sum for each grid C_{ij}
Output: Extracted point cloud as new individual edges
  for each column of the grid **do**
  find the convolution sum of the left C_{j-1} and right C_{j+1} neighbouring columns
  if $C_{j-1} = 1$ and $C_{j+1} = -1$ and $C_j = 0$
  extract the centroid corresponding to grid C_j as the new edge E
  if $C_{j-1} = 1$ and $C_{j+1} = -1$ and $C_j = 1$
  the left boundary of the thick edge, extract the centroid of grid C_j as E_l
  if $C_{j-1} = 1$ and $C_{j+1} = -1$ and $C_j = -1$
  the right boundary of the thick edge, extract the centroid of grid C_j as E_r
  Calculate and extract new edge as $E = (E_l + E_r)/2$
  Take E as the new extracted edges
End

Table 1
Edge detection result corresponding to different boundary types.

Convolution sum	1	−1	5	−5	others
Boundary Type	left edge	right edge	upper edge	lower edge	noise

3.3.3. Extraction of bridge cross-section parameters

The bridge cross-section has non-standard geometry that involves multiple parameters which require higher accuracy. Extracting these parameters is also constrained by the complexity of the cross-section and the quality of the point cloud. This increases the complexity of the edge extraction; hence it is necessary to improve the mesh delineation method as discussed in previous section.

As shown in the point cloud in Fig. 4, the resulting edge point cloud appears as two thick layers, which increases the width of the contour and the complexity of the shape. To address this challenge, this study proposes a method based on the combination of gridding and edge operator detection for bridge contour edge detection.

In this paper we propose a new method for point cloud edge detection, inspired by the gradient-based edge detection method, where the point cloud is meshed and then edge extraction is performed using an edge detection operator. The core innovation of this method lies in the improvement of the boundary extraction operator, enabling its direct application to layered, low-quality point clouds. As shown in Fig. 4, the operator is applied to the point cloud to obtain the theoretical boundary, represented by the red dashed lines, which enhances the quality of the point cloud and improves accuracy.

As shown in Fig. 9, for the cross-section slice of the bridge, which itself is a rough and layered point cloud contour, if the traditional operator is applied directly, the whole cross-section will be extracted, and the threshold for distinguishing the edge is difficult to be determined. Thus, the objectives of the method should be, firstly, to break the whole contour into different edges and distinguish the left and right layers of each edge of the contour; secondly, to extract the middle of the two layers to generate a new edge. In the following, the use of the proposed improved operator for both operations are explained.

Firstly, the point cloud is assigned to a grid with a size of w following Eq. (3). A histogram-based method is used to determine the grid width w. The sliced point cloud is divided into bins based on the column arrangement, with the bin width corresponding to the precision unit of the edge width. Since the grid is square, the grid size w is determined by vertical edges only (vertical edges are most prone to layering). The number of points in each bin is displayed in the form of a histogram. Taking the point cloud in Fig. 9 as an example, the corresponding histogram is shown in Fig. 10.

Since the aim is to divide the thick edges (shown in Fig. 4) into two layers (left and right), the grid width w is set to half of the average width of all edges, see Eq. (3). Assuming the vertical boundary widths are $w_1 \ldots w_6$ and n represents the number of vertical edges.

$$w = \frac{1}{2n} \times \sum_{i=1}^{n} w_i \tag{3}$$

Then, the convolution sum is computed for each grid (see Fig. 11), the convolution sum c represents the summation of each element R_{ij} in the resulting matrix $\mathbf{R}$. Taking the grid to be computed (marked with a red star) as the centre, a 3 × 3 sub-grid is formed with eight grids in the nearby neighbourhood. Firstly, the weight matrix of the sub-grid is found and assigned a value of 1 if there is a point cloud in the grid and 0 if there is no point cloud in the grid. Secondly, this weight matrix and the modified operator (convolution kernel) are used to do the convolution operation, and finally each element of the resulting matrix is summed up to get the convolution result of each grid.

For a certain column grid, if the values of the grid are all 1, it means that there is a column of point on the right side of this column, which is the left boundary; if the value of the column grid is −1, it means that there is a column of point on the left side of this column, which is the right boundary. When a 3 × 3 convolution kernel (weight matrix) is convolved from left to right with a stride of 1, the convolution has two possible results (shown in Fig. 12), the results of different convolutions have been highlighted in different colours.

Firstly, for layered edges, the two adjacent layers are divided into neighbouring left and right grid columns, where the convolution sum of the left column is 1 and that of the right column is −1. In this case it is only necessary to take the middle value for the point cloud of the two

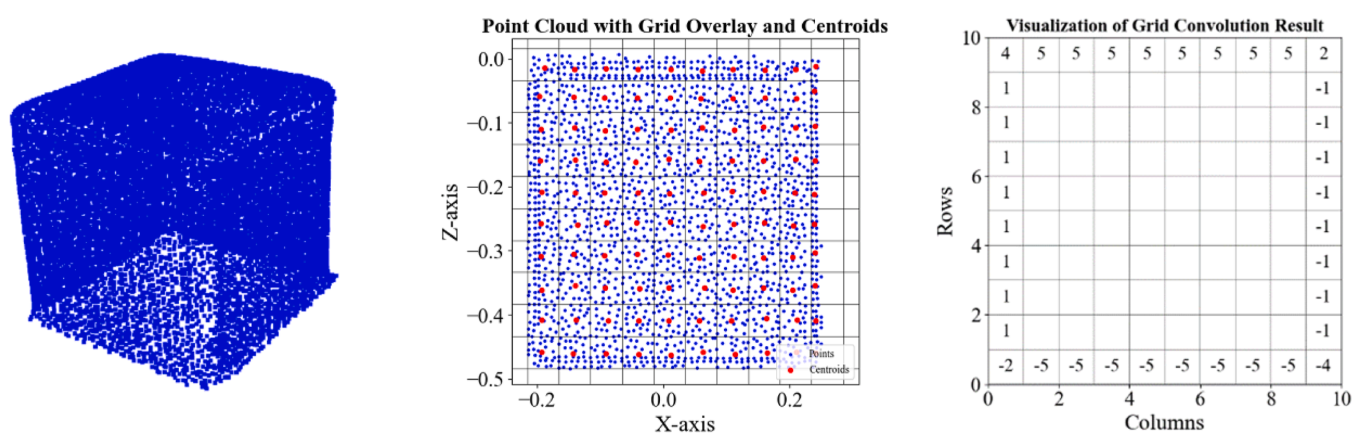


Fig. 13. Edge detection of concrete cube.

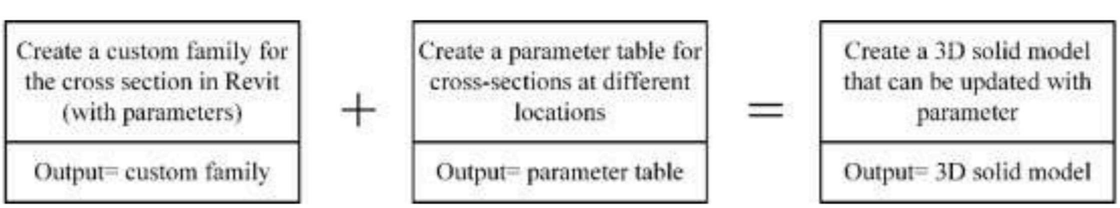


Fig. 14. Implementing parameterisation and updatability of models.

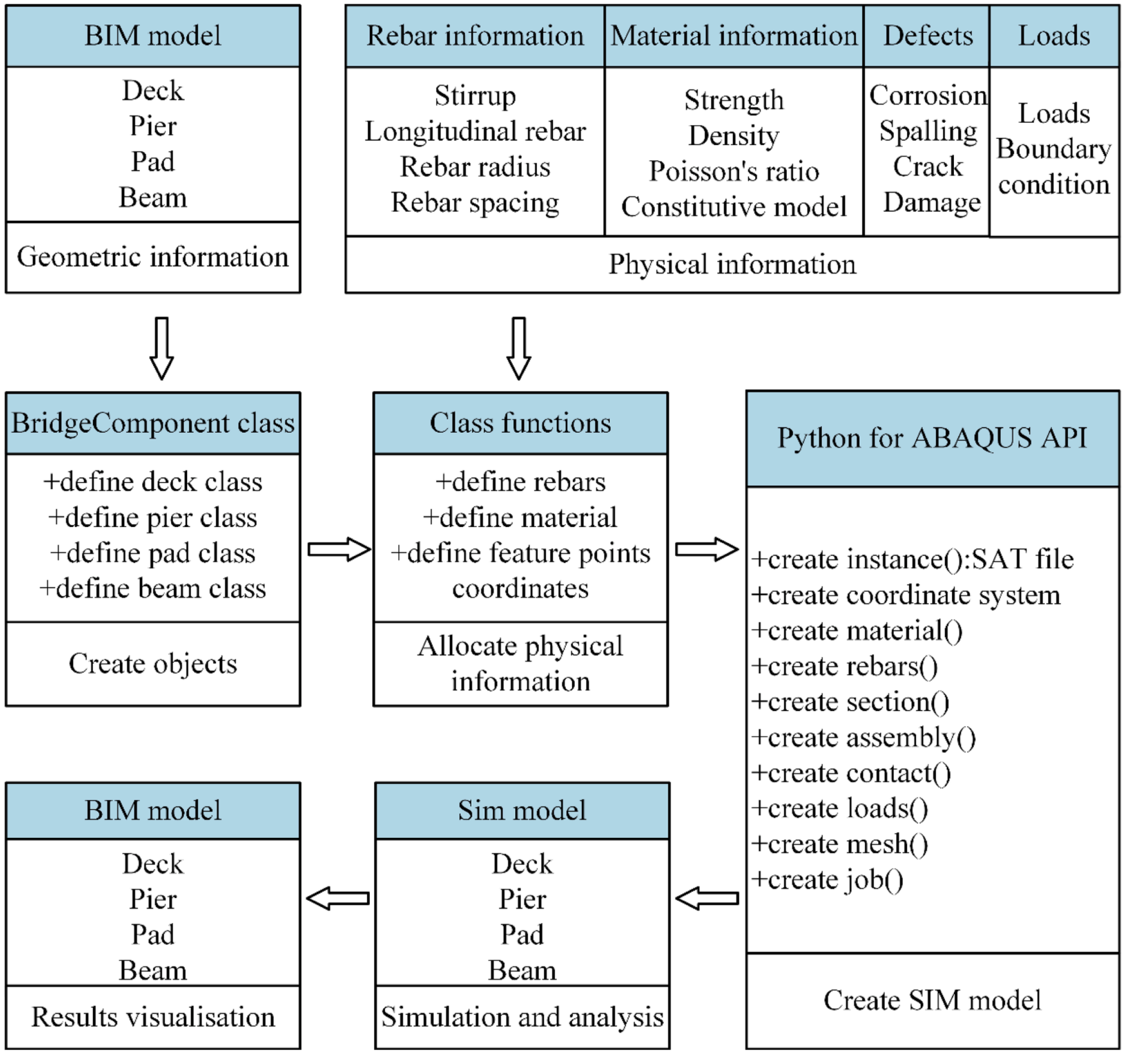


Fig. 15. BIM to Simulation framework.

columns as the new edge. The second result is that, for non-layered edges, the boundary is divided into one column of the grid. The convolution sum of this column is 0, and the sums of the adjacent left and right columns are 1 and −1, respectively. Thus, the point cloud within the grid of this column is directly extracted as the new edge. The major steps of the new edge extraction module are presented in Alg. 1.

As shown in Alg. 1, C_{ij} represents the convolution sum for each grid, where i and j denote the row and column numbers, respectively. E_l and E_r indicate the left and right boundaries extracted based on the convolution results.

Similarly, this method can be used to differentiate between up and down boundaries. Therefore, in this method, only the values of ±1 and ±5 need to be considered, while the remaining values are regarded as irrelevant points, which greatly reduces the number of computations. Thus, the left, right and top, bottom edges can be distinguished directly from the convolution of the grid. For layered edges, the convolution results corresponding to different boundary types are shown in Table 1.

Different types of boundaries are divided according to the different results, and there is no need to use multiple operators for detection, which improves the convenience of calculation. The method can also be used for high-quality point cloud edge detection. For a uniform point cloud with clear edges, our operator remains applicable, with the mesh size adjustable to the average edge width.

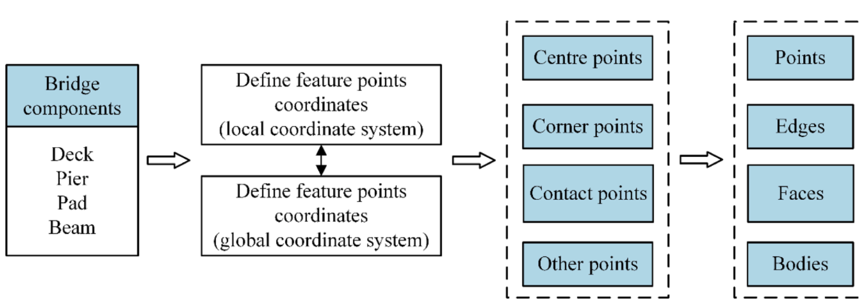


Fig. 16. Methods for adding coordinate attributes to bridge components.

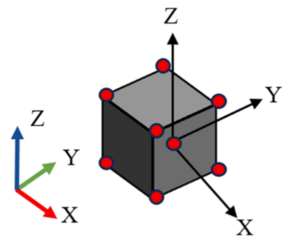


Fig. 17. Feature points of pad.

To demonstrate the generality of this method, a concrete cube is taken as an example, and the contour detection is schematised as illustrated in Fig. 13.

The point cloud of this cube is uniform and does not exhibit layering. However, by using the proposed operator, the boundaries can be accurately distinguished based on the convolution results, which demonstrates the general applicability of our method.

After successfully extracting the point cloud of different edges of the cross-section, a primary function is used to fit the different point cloud contours to get the intersection points of the fitted function as the corner points of the contour, and the horizontal and vertical distances between the corner points are found to be the parameters of the cross-section of the bridge.

3.4. Generation of 3D model of bridge

Using extracting the parameter information of the bridge, the 3D model is reconstructed in the BIM software according to these parameters. In this paper, a 3D model is generated by firstly defining the centreline of the bridge, then producing the centre point of the sections on the centreline according to the pre-set interval and extracted elevation, and finally placing the cross section on the centre point. The cross section is created according to the extracted parameters, and finally the solid model is obtained by connecting the two cross-sections to generate the solid, as shown in Fig. 2.

This step is carried out using the Dynamo programming software (Dynamo Community, 2023) that is used to automate model generation in BIM software Revit. The cross sections are drawn using a predefined family of cross sections. To form the digital model, the instances of the family are imported with customised parameters that change with position as variable parameters (see Fig. 14).

This change of the parameter values of the cross sections is realised through the "set element by parameters" node that is implemented to change the cross-section shape.

3.5. BIM and simulation software integration

It is necessary to convert real-time 3D models into numerical simulation models for structural health diagnostics and future condition predictions. Bridge information includes both geometric information (such as the shape of the bridge) and physical information (such as reinforcement, loads, materials, and defects). Geometric information is imported into numerical simulation software through BIM models, whereas physical information relies on the engineer's manual operations to input complex information into the numerical analysis software. This process is time-consuming and labour-intensive and requires familiarity with the software. Therefore, a BIM-to-Simulation workflow is required to automate the representation and analysis of the structure.

In the BIM-to-Sim section, geometric information and physical information are imported into the simulation software. The former is imported via SAT files, while the latter is imported using Python scripts developed in this study (see Fig. 15). The Python scripts serve as an information storage and processing library, automatically assigning physical properties to the various components of the bridge. The script utilises both, BIM model and user input, providing the necessary information to the simulation software based on its requirements. Traditional simulation analysis typically involves significant manual input and interaction within the software, which not only increases the workload but also introduces the potential for errors. The Python script serves as an interface that maximises automation and performs input data validation. This is implemented by incorporating conditional statements in the code to ensure that the input parameters for the constitutive model are reasonable, such as requiring the plastic strain to be greater than zero. This prevents calculation errors caused by incorrect data in the simulation, significantly reducing the manual workload. Lastly, the simulation results is extracted for visualisation in the BIM model.

The specific operation involves leveraging the object-oriented programming features of Python by defining bridge categories as separate Python classes, with each bridge component represented as an instantiated object of these classes. The process of adding various physical properties to the bridge components is implemented through internal functions defined within the classes. By incorporating appropriate numerical logic within these functions, the script performs data validation to ensure the input data's correctness. Our Python script is highly compatible with numerical software and works seamlessly with the Python API in ABAQUS, facilitating the extraction and transfer of data for numerical analysis.

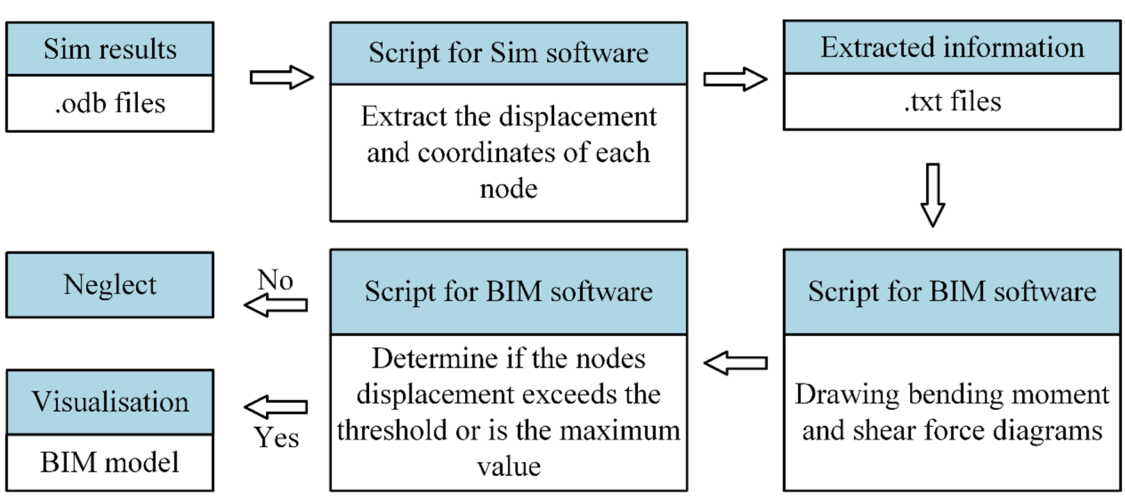


Fig. 18. Sim to BIM framework.

Fig. 19. Location of the bridge (C3W6+RX Birmingham) and collection points (a) and photo of the bridge-Ariel Aqueduct (b).

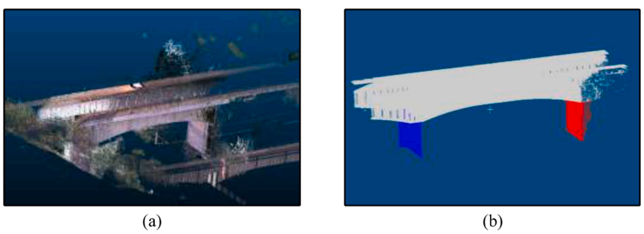


Fig. 20. Raw point cloud data (a) and processed point cloud data (b).

3.5.1. *From BIM to simulation*

In our Python script, four Python classes are defined according to the classification of bridge parts: deck, beam, pier, and pad, and objects of each class are created according to the bridge component. In each class, attributes such as rebar, material properties, and feature points are assigned to the component object through the definition of class functions. Constraints, loads, contacts, and other elements are then incorporated into the model via the ABAQUS Python interface.

Simulation modelling often involves applying forces and constraints to specific geometric regions. If the geometry of the bridge is complex, accurately interacting with the geometric region through code may be challenging. To address this, the findAT() function is utilised to locate the specific area. This function is a built-in Python method for ABAQUS, designed to identify points, lines, surfaces, or bodies that intersect with the provided coordinates. The input consists of one or more coordinates, while the output is a set of entities (points, edges, faces, or bodies) that pass through those coordinates. Consequently, a method must be developed to identify coordinates that uniquely define the region.

To address the problem, this paper proposes the following solution. A new attribute about coordinates is added for each component (see Fig. 16). This attribute stores the coordinates of the feature points for each component, which uniquely define the component shape. By combining feature points, the corresponding geometric entities can be obtained. Two feature points can determine an edge, and three or more feature points can define a surface or a body. It is implemented by creating a class function in each bridge class using Python.

Each component should have two sets of feature points (see Fig. 17), corresponding to the coordinates in the global coordinate system and in the local coordinate system. The purpose of this is to be able to import and analyse all the parts into the finite element software (use global coordinate system) or to analyse them individually (use local coordinate system) for subsequent analyses.

By defining the feature point, the corresponding coordinates that uniquely identify the specified region can be obtained. These coordinates are then used as input for the findAT() function. Subsequently, boundary conditions and interactions are defined by assigning the result of function to sets.

3.5.2. *From simulation to BIM*

After obtaining the analysis results from the Sim software, to analyse the results more intuitively, it is necessary to map the results from the simulation software to the BIM software for visualisation. This step aims to enhance data integration and visualisation. As a comprehensive information model, BIM enriched with real-time structural analysis results further expands its semantic information, reducing reliance on additional software. Moreover, this approach enables the visualisation of structural responses of the bridge to different loads, supporting effective lifecycle management. As shown in Fig. 18, to achieve this step, Python for ABAQUS is utilised to post-process the .odb result files, extracting stress at the model nodes and deformation at the integration points,

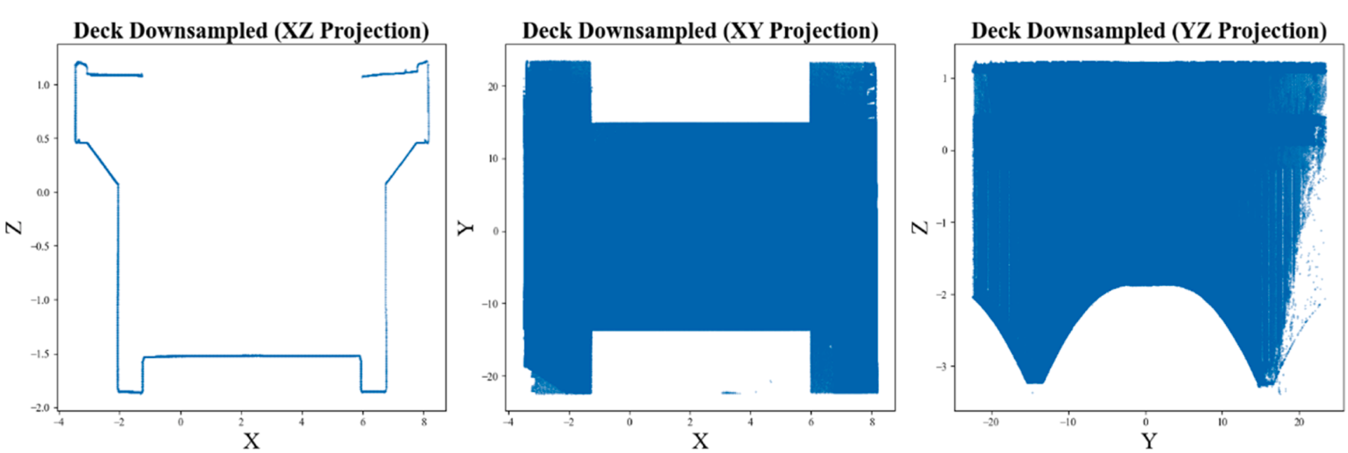


Fig. 21. Projections of Bridge Point Cloud Data.

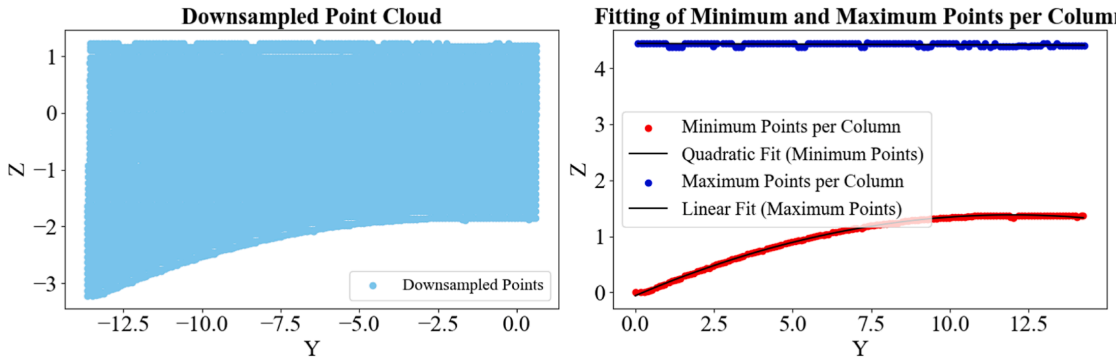


Fig. 22. Quarter-YZ Plane Projection of Deck and Fitting Results.

along with their corresponding coordinates.

Subsequently, Dynamo scripts are employed to further process the extracted stress and strain, filtering out deformation that exceed predetermined thresholds. Points are then drawn within the BIM model according to their coordinates.

4. Case study

The case study bridge is located near the south gate of the University of Birmingham and is a single span girder bridge approximately 55 m long and 12 m wide. The bridge is in service, crossing the A38 as part of the Worcester & Birmingham Canal, with the canal channel in the centre for the traffic of vessels, with pedestrian crossings on both sides of the bridge, the location and shape of the bridge is shown below.

4.1. Scan-BIM practice

In this study, a Trimble SX10 scanning total station is used to collect the point cloud data. The scanner is technically suitable for the task of collection of data with high resolution point cloud at acceptable speed. The collection points are set up (see Fig. 19(a)), and the point cloud of the bridge is acquired using the methods mentioned in Section 3.1. The Cloud Compare software is used to de-noise the bridge point cloud by applying the SOR and CSF methods. Based on the characteristic of a sudden change in the quantity of point cloud data at the bridge pier, the point cloud is segmented vertically using bounding boxes to separate the bridge deck. Subsequently, the point cloud is clustered using k-means based on the normal vector and coordinates to separate the bridge pier. The processed bridge point cloud is shown in Fig. 20(b)).

The rotation angle determined by the PCA method is 12.285°. The piers are of the combined type and the parameters extracted are R = 1000 mm, L = 7000 mm, H = 6000 mm. Due to the issues with the quality of the point cloud in this case, the accuracy of the reconstruction is controlled to 10 mm, which is acceptable for the reconstruction purpose. Therefore, after rotating the bridge point cloud, a 3D mesh in size of 10 mm is used to down sample the point cloud. The voxel down-sampled point cloud initially contained 7963,354 points, which is reduced to 4251,450 points after down-sampling, achieving a size reduction of 46.6 %. The projection after rotation and down sampling is shown in Fig. 21:

The longitudinal section parameters are determined from the xz projection. Due to obstructions in line of sight from shrubs on both sides, there are gaps in point clouds on both sides of the bridge. Since the bridge is symmetric, it is folded, and a quarter of the bridge is taken to extract the point cloud. Due to the redundancy of the point cloud of the bridges, the voxel down-sampling of the bridges is carried out before applying grid segmentation. The grid size used is 0.1 m, which can effectively balance computational load and accuracy for a long bridge. The points with the maximum and minimum values of z-coordinate on each grid column are extracted to get the boundary point cloud respectively. The longitudinal section equation of the bridge can be obtained through the fitting results as:

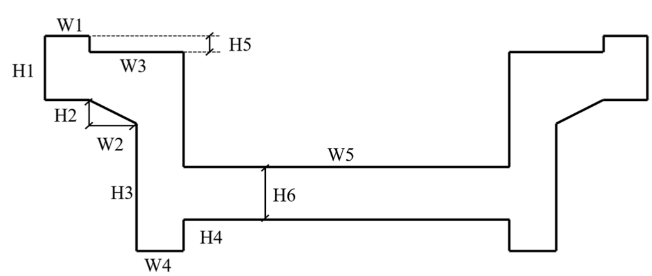


Fig. 24. Parameters of the deck.

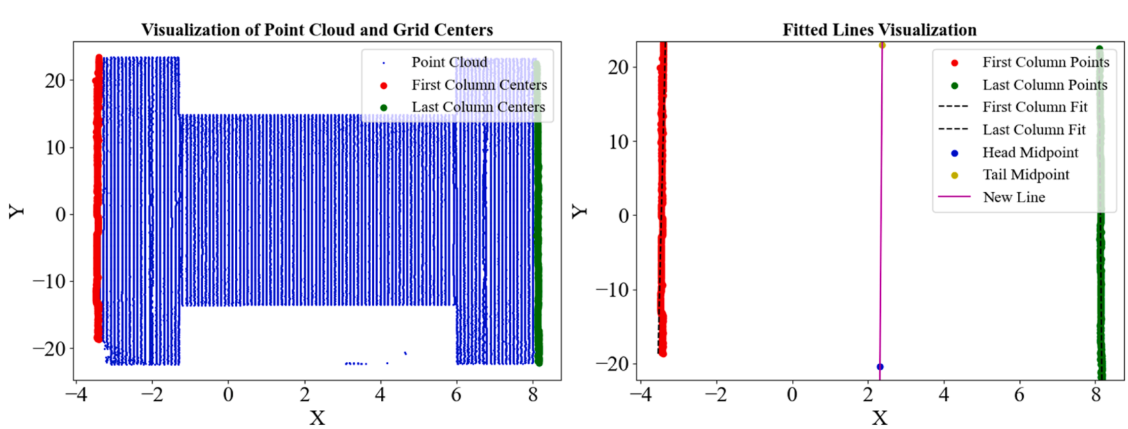


Fig. 23. XY Plane projection of deck and fitting results.

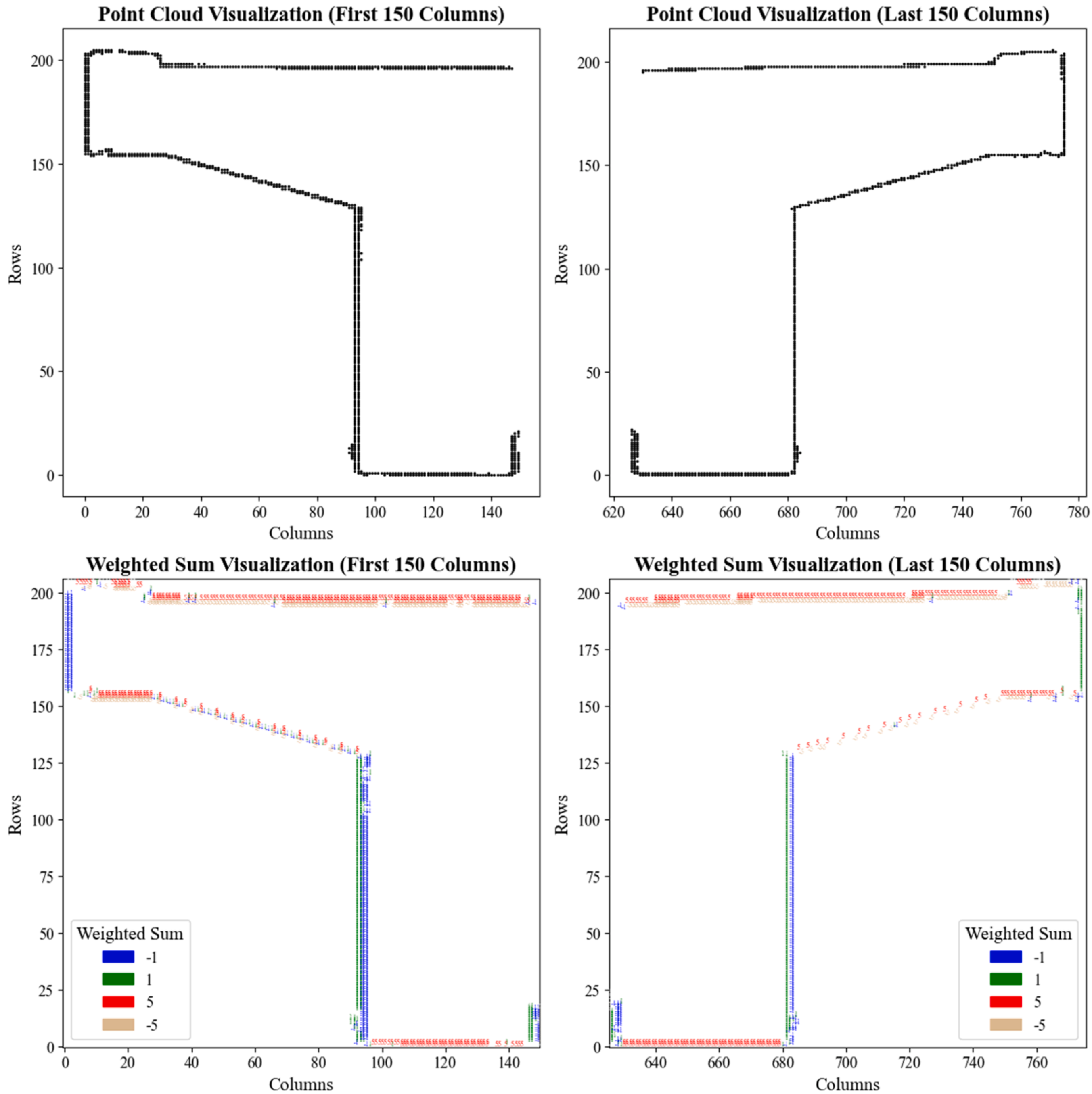


Fig. 25. Results of bridge deck edge detection.

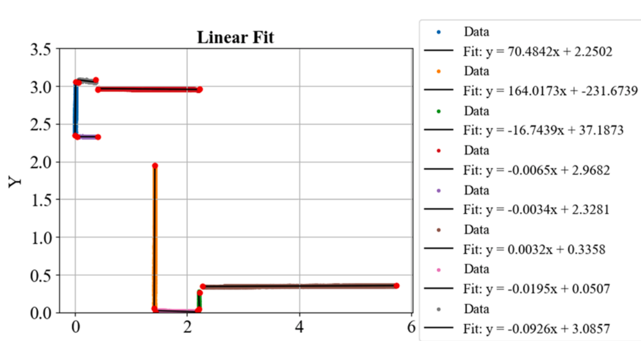


Fig. 26. Result of linear fitting of each edge.

$$z = 0.0004x + 4.3755(m) \tag{5}$$

Since the slope is 0.0004 (the elevation change is <0.4 mm), it is considered that there is no longitudinal slope parameter of the bridge, and it can be approximated that the bridge is horizontal, meanwhile, the girder height of the bridge shows a quadratic curve, where x, y and z are coordinates of the point cloud.

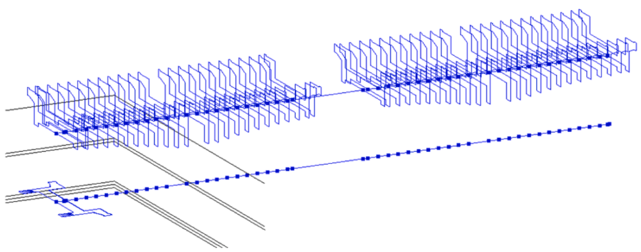
$$z = -0.011y^2 + 0.243y - 0.011(m) \tag{6}$$

Hence, the girder height at any position can be obtained by this quadratic curve. The illustration of the obtained parameters is shown in Fig. 22.

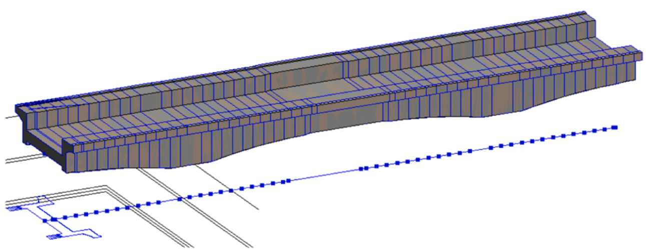
The bridge centreline parameters are subsequently determined from the xy projection of the bridge. Similarly, the projected point cloud is

Table 2
Table of deck cross-section parameters (see Fig. 24).

Parameters	H1	H2	H3	H4	H5	H6	W1	W2	W3	W4	W5
Value (mm)	750	400	1950	350	50	1000	400	1000	1800	800	7150



a) Drawn centreline and cross sections



b) Drawn deck (after swept blend operation)

Fig. 27. Drawn centreline and cross sections a) and reconstructed bridge deck.

divided using a grid with a side length of 0.1m, and the points in the leftmost and rightmost grids are taken as boundaries, as shown in Fig. 23. As the bridge is a linear simply supported girder bridge, the centreline of the bridge is a straight line, the equation of the centreline is

$$y = 727.8352x - 1706.4254(m) \tag{8}$$

and the coordinates of the start and end points of the bridge are (2.3164, −20.4424) and (2.3760, 22.9322).

Fig. 23

The cross section of the bridge is more complex with multiple vertical and horizontal boundaries. Thus, the cross section is parameterised as in Fig. 24.

According to equation (4), the average width of the six vertical boundaries is obtained as 25 mm, so a 15 mm grid size is determined for cross-section edges extraction. Convolution results of the grids are obtained by the method proposed in Section 3.3.3, and the visualisation results are shown in Fig. 25 (edges are represented by different colours), then the horizontal and vertical boundary lines are successfully distinguished and extracted.

The method is effective in extracting and classifying all boundaries in both horizontal and vertical directions, and the longer the line segments are, the better the extraction results which are achieved. Boundaries with a certain slope cannot be extracted directly by columns or rows. Nevertheless, they can be extracted by excluding horizontal and vertical meshes. The extracted boundary point cloud is fitted with straight lines, and the intersection of the fitted lines and the endpoints of the fitted line segments are obtained. The extracted and fitting results are illustrated in Fig. 26.

The corner points can thus be determined by the endpoints of the fitted line. Then the parameters of the cross section are obtained by finding the distance between corner points. Since the deck is obscured by the canal, the deck thickness $H6$ cannot be directly obtained from the scanned data. In this case, it is assigned a value of 1000 mm. The obtained results following rounding are shown in Table 2.

After the cross section generated from the parameters (see Fig. 27 Drawn centreline and cross sections a) and reconstructed bridge deck. a) (the lower line shows the XY-projection of the centre line and the upper line shows the line after translating its elevation upwards), reconstructed 3D models of bridge decks can be successfully generated (see Fig. 27 Drawn centreline and cross sections a) and reconstructed bridge deck. b).

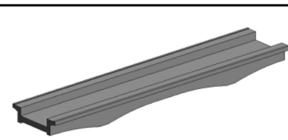
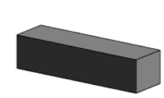
The 3D model of the remaining members, tie girders and pad stones can be obtained in the same way. All bridge components and the complete BIM model are shown in Table 3 and the whole bridge is assembled to get the overall 3D solid model (see Fig. 28).

4.2. BIM-Sim integration

Furthermore, the BIM-to-Sim workflow is employed to assess the structural performance of the bridge. According to the BIM-to-Sim method described earlier, the geometric information of the bridge is imported into ABAQUS in the SAT format from the BIM model. The physical information is imported into ABAQUS using the script proposed in Section 3.5. The physical information in the Python script for ABAQUS includes rebar distribution, prestressing arrangement, material properties, load distribution, and coordinates for boundary condition placement. These details are defined and assigned to the bridge components within the script, then automatically extracted and input into ABAQUS, reducing manual operations and saving time. Additionally, the information is verified before being input into ABAQUS to prevent analysis errors caused by incorrect input.

In the absence of reinforcement drawings of concrete sections, rebar information can be defined by setting the appropriate parameters. These parameters have engineering significance and support subsequent changes. In this paper, the corresponding reinforcement parameters are set as reinforcement rate ρ, thickness of reinforcement protection layer c, and reinforcement type and diameter d. The insertion of reinforcement is done by inserting surface units in ABAQUS to simulate the rein-

Table 3
3D model of Bridge Components.

Component	Deck	Pier	Pad	Tie Grider
3D Model				

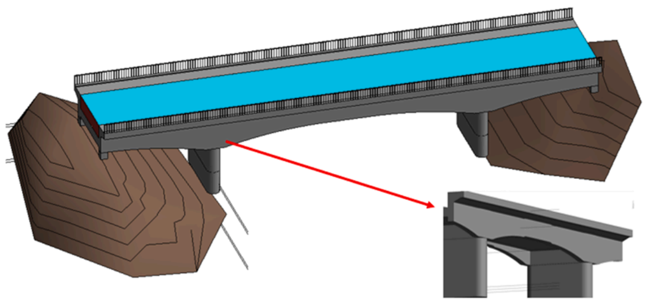


Fig. 28. Complete BIM model and detail (bottom right).

forcement layer. For the purpose of analysis in this paper, the reinforcement rate is set to 1 % [67,68] and the rebar diameter to 25 mm (these are parameterised in the code and support modification). Based on the cross-sectional area the rebar area can be obtained, thus introducing the rebar spacing, which serves as input for setting the rebar layer.

In this paper, the Concrete Damage Plasticity (CDP) [69] model is employed to simulate the non-linear behaviour of the reinforced concrete. In this case study, the materials used are C30 concrete and reinforcing steel with a yield strength of 500 MPa and the corresponding stress-strain curves of concrete and rebars are shown in Appendix 1 and Appendix 2. The other plastic parameters are determined from [70–73] with the dilation angle $\varphi = 31^{\circ}$, the flow potential eccentricity $e = 0.1$, $f_{b0}/f_{c0} = 1.16$, $k = 0.667$ and the viscosity parameter $c = 0.0001$. Regarding the loads and boundary conditions, uniform loads q_e is applied to simulate the prestressing tendons using the force equilibrium method [74–76]. Based on Eurocode 1 [77] and reasonable assumptions, the permanent load (gravity G and water pressure q_w (15.7 kPa) are applied in the force balancing method, while the loads applied in the force analysis are pedestrian load q_p (5 kPa) and boat load q_b (1.96 kPa) on the bridge (see Fig. 29). The load determination process is detailed in Appendix 3.

The case is a continuous bridge, according to the photo (see Fig. 19 (b)) it can be seen that the piers are fixed connections, and the rest of the bearings are placed on 80 mm square pads which are located on the four corners of the bridge. As shown in Fig. 30, the bearings limit the displacement of the bridge in the transverse direction (x-direction) and allow the bridge to move in the longitudinal direction (y-direction).

Due to the complex shape of the bridge, a mesh size of 380 mm is used to simplify the meshing step. To define a more refined mesh, a partition can be created manually in the parts of the bridge where the shape changes. The BIM model is converted to a FEM by following the workflow presented in Section 3.4. and information on reinforcement, material loads and constraints is added. Then, Python scripts are used to automate the model development and job submission processes, and the simulation is completed in ABAQUS. Based on the force equilibrium method [78–80], the applied prestress is determined using permanent loads. The shear force and bending moment diagrams of the bridge under permanent loads are shown in Fig. 32(a) and (b). The displacements under different loads application and the stresses in the vertical direction are illustrated in Fig. 31.

From the results, the maximum deformation of the bridge is approximately 3.8 mm under the permanent load (see Fig. 31(a)). After applying the prestressing force, the bridge has a certain anti-arch when it is not subjected to the traffic load (see Fig. 31(b)), which indicates that the prestressing force is applied effectively. After applying the prestressing load, the boat and pedestrian loads are applied as dynamic loads to the middle and both sides of the bridge, results are shown in Fig. 31(c).

Following the method in Section 3.5.2, the FEM results are presented in the BIM model. When the settlement value at a given point exceeds the set threshold, it is visualised as a blue point in the BIM model. As

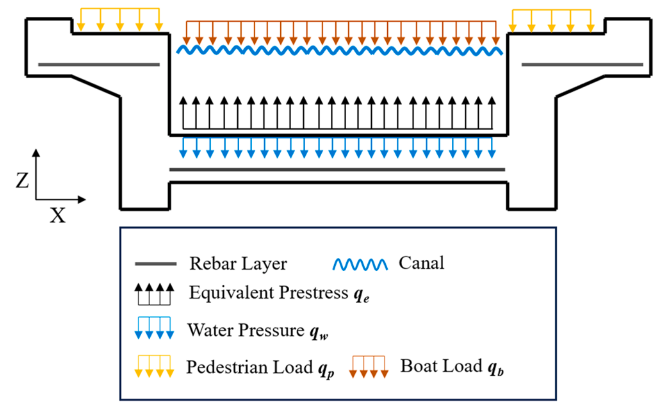


Fig. 29. All applied loads on ABAQUS model.

shown in Fig. 32(c), the blue region gradually narrows as the threshold changes from 0.3 mm to 0 .5mm. And the shear force diagrams and bending moment diagrams at the corresponding locations of the bridge are visualised in the BIM model.

4.3. Performance evaluation

To evaluate the point cloud segmentation step, the manually segmented reference point cloud is compared with the point cloud segmented by our method. Precision and Recall are used to measure segmentation accuracy, with values of 0.9295 and 0.9294, respectively, indicating a high consistency between the segmented point cloud and the manually segmented point cloud. Due to the absence of bridge drawings, this study validates the reconstruction results by comparing the measured bridge components with the reconstructed BIM model. The same laser scanner (Trimble SX10) is utilised to collect reference point clouds. To minimise errors caused by scanner movement and point cloud registration, the laser scanner remains stationary during the measurements, capturing data from a single station only. The scanner is positioned beneath the bridge to acquire reference point clouds of the piers, as shown in Fig. 33. The bridge parameter L, pier parameters w4 and w5, and the distance between the two piers are used for validation. The average error obtained is 0.01 m, indicating a high level of accuracy in the reconstruction achieved using our method.

In the Scan-to-BIM-to-Sim framework, the process of extracting parameters from the scanned point cloud occurs instantaneously, with negligible waiting time. The speed at which the bridge BIM model is generated in Revit is influenced by the CPU model and the number of cross-sectional slices. Specifically, employing an Intel® Core™

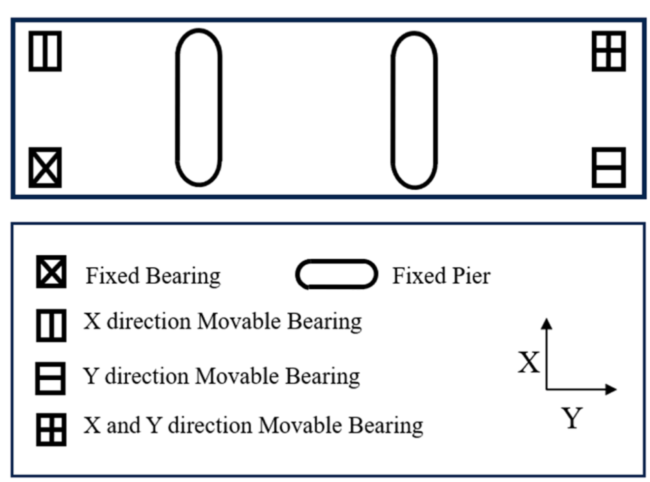
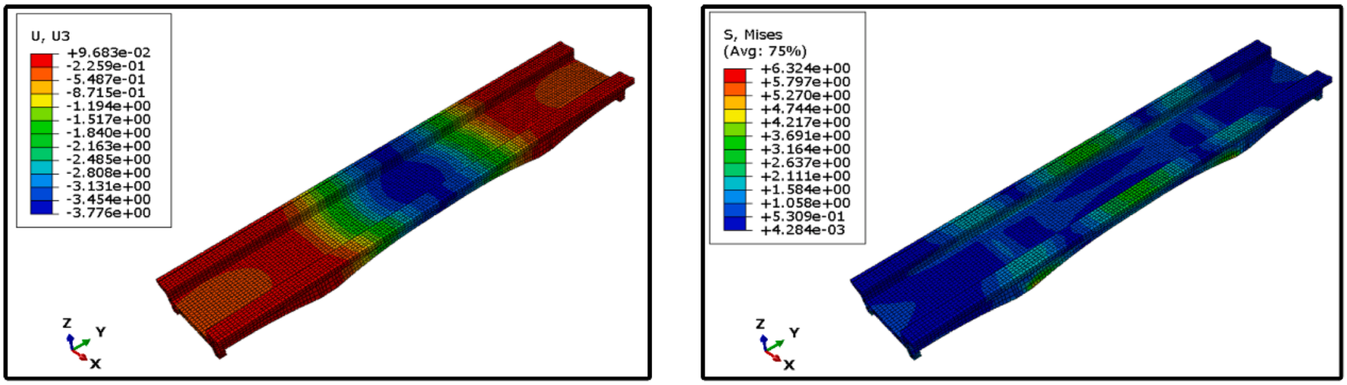
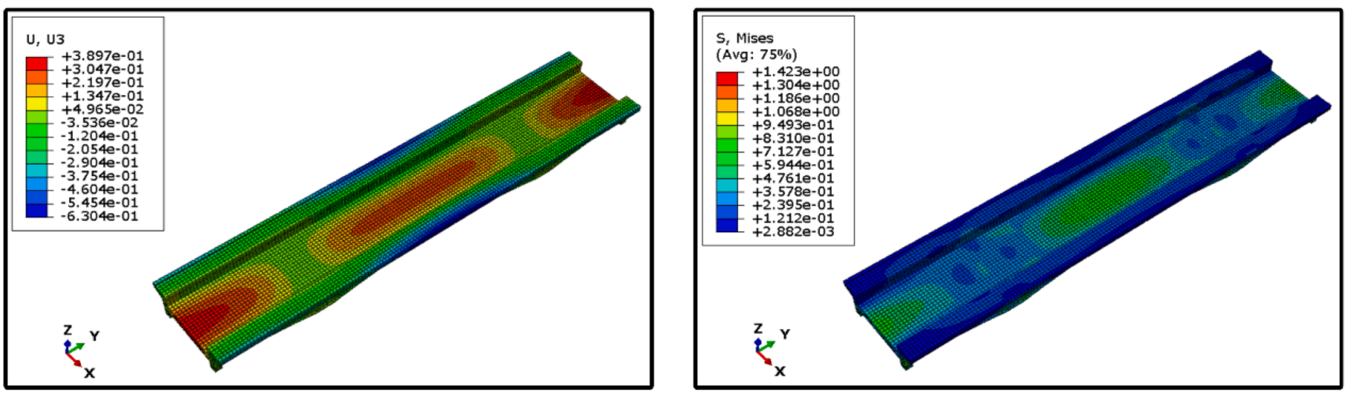


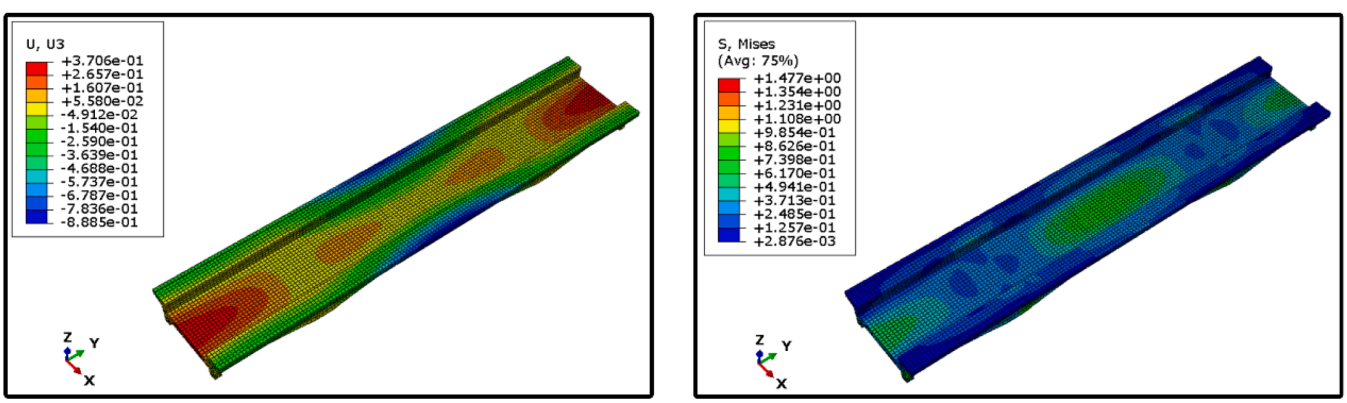
Fig. 30. Boundary conditions of the model.



(a) Vertical displacements (mm) and Von Mises stress (MPa) under permanent load (before prestress).



(b) Vertical displacements (mm) and Von Mises stress (MPa) under permanent load (with prestress).



(c) Vertical displacements (mm) and Von Mises stress (MPa) (under permanent, prestress and live loads).

Fig. 31. ABAQUS results under different load combinations.

i7–9750H CPU at 2.60 GHz with 100 cross-sectional slices to generate 11 parameters, the process requires approximately two minutes. The bidirectional mapping between the BIM model and the simulation model is instantaneous. However, the duration required to obtain simulation results is contingent upon the analysis time in the numerical software. In the case study, the period from submitting the analysis task in ABAQUS to its completion is ten minutes. This demonstrates that the framework proposed in this paper effectively reflects results and updates in real time, thereby underscoring its practical applicability.

5. Discussion

The framework proposed in this paper fully automates the Scan-to-BIM-to-Sim process, addressing the gap in a holistic and comprehensive approach that not only reconstructs digital models but also integrates automation in assessment, enabling improved diagnostics and virtual stress testing. Compared to previous studies based on high-quality point clouds, this research introduces a solution also for imperfect data, reducing reliance on equipment during data collection and offering a solution for model reconstruction with lower-precision scanning devices, thereby significantly reducing costs. In the BIM-to-Sim workflow, this study we propose systematic and robust link between BIM and simulation software, automating the importation of both geometric and physical information into the numerical model, completing the analysis, and mapping the results back to the BIM model, thus enabling the retrieval, reuse and flow of bridge information.

However, the reconstruction of the BIM model in this study does not include the complex surface decorations and chamfers of the bridge. In

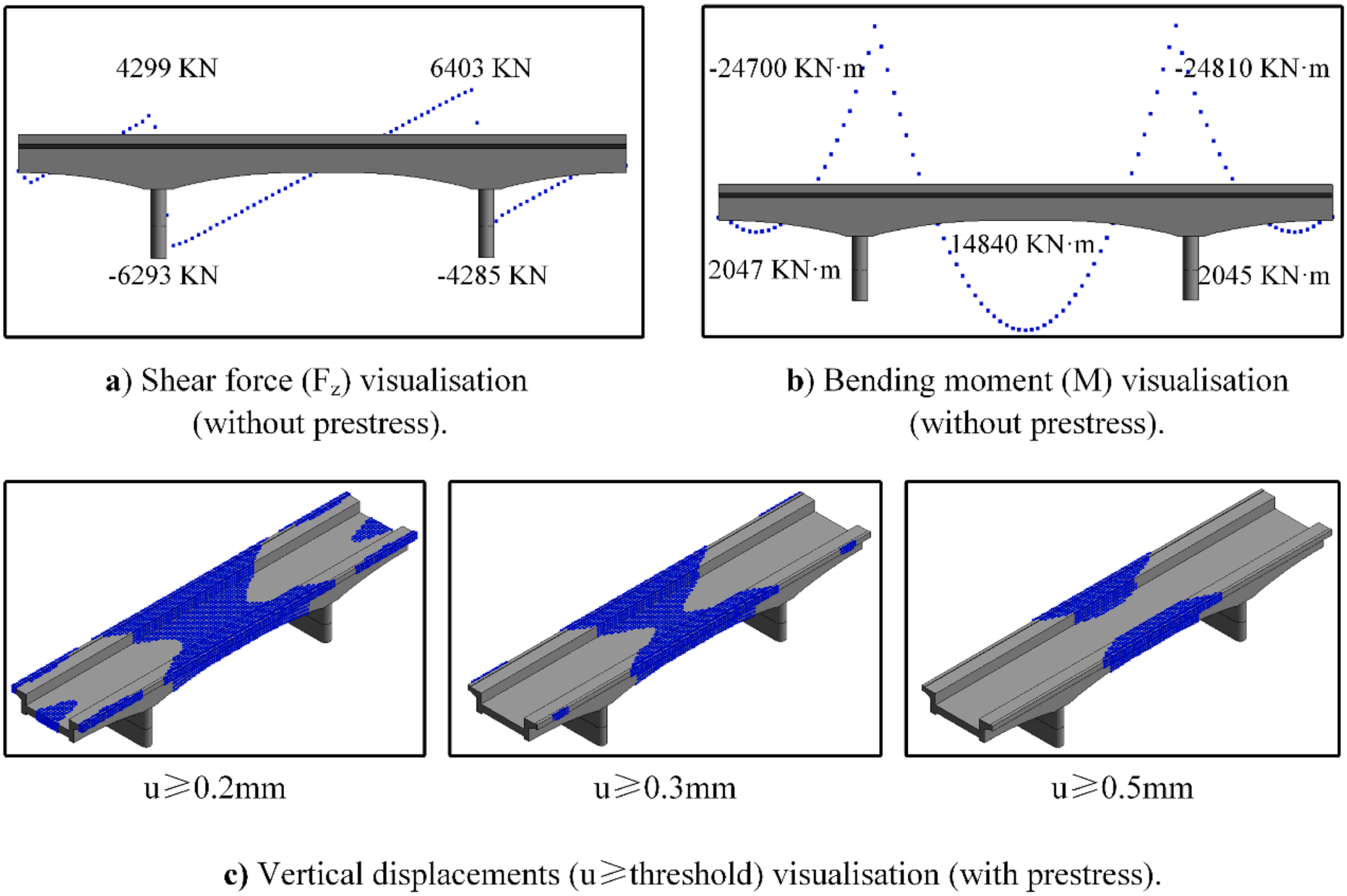


Fig. 32. FEM results visualisation.

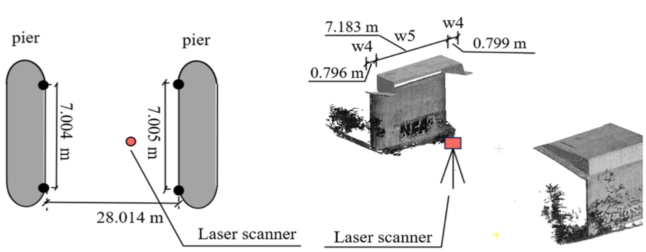


Fig. 33. Measurement results and the scanned reference point cloud.

the point cloud segmentation process, a clustering method is employed for automated segmentation. While this approach is effective for simple concrete bridges, its accuracy decreases for bridges with a greater variety of component categories. Future research could explore more precise techniques, such as the use of pre-trained large vision models for point cloud segmentation without training requirements, or diffusion models for point cloud completion. In the BIM-to-Sim process, the work presented in this paper is limited to reinforced concrete bridges in good condition, hence not containing information on defects of the bridge. Future research could explore integrating the health conditions of bridges, such as spalling, corrosion, and cracking, into the Sim analysis. Moreover, future research could focus on importing physical data with varying levels of detail into simulation software based on specific analysis requirements. Additionally, it is crucial to consider the effects of prestressing and the structural integrity of connections, bearings, expansion joints, and other supplementary devices. Addressing these elements is essential for generalising the method to accommodate all types of bridges, ensuring that the simulation accounts for the full spectrum of structural dynamics and potential degradation over time. This comprehensive approach would enhance the predictive accuracy and reliability of bridge assessments.

6. Conclusions

Current BIM-based reconstruction frameworks for bridges require substantial manual effort, rely heavily on precise equipment, and demand significant computational resources, all of which contribute to high costs. To address these challenges, this paper proposes a consistent Scan-to-BIM-to-Sim framework that enables automated digital model reconstruction and efficient analysis. This approach is precise, computationally efficient, and requires minimal manual intervention, thereby significantly reducing costs. The main idea is to use custom Python scripts as a bridge, enabling data transformation, interaction, and integration between scanned data, BIM models, and Sim models.

The novelty of the proposed framework lies primarily in its robust point cloud edge detection method based on meshing and detectors, which can be directly applied to rough, low-quality point clouds to achieve precise results. This method effectively classifies and extracts edges (top, bottom, left, and right contours) of the point cloud using an edge detection operator. More importantly, it reduces dependence on costly equipment for 3D model generation, thus lowering overall expenses. By addressing these data quality challenges, the aim is to enhance the robustness and applicability of point cloud processing methods, ensuring that method can effectively handle and correct such irregularities. This approach contributes to the field by broadening the scope of point cloud data processing applications, while also ensuring that data collected under suboptimal conditions remains valuable and usable.

Furthermore, this paper improves the framework for automated bridge reconstruction from point cloud to 3D model to finite element model, introducing an innovative interaction between code and the ABAQUS interface through automated creation of Python classes and feature points for each component. This improvement facilitates data interaction and mapping across raw data, BIM models, and simulation models, enhancing the framework's automation and efficiency. Additionally, the framework maps numerical analysis results onto the BIM model, allowing real-time visualisation and enriching BIM semantic information. The framework supports diverse scenarios for simulation and structural analysis. By comparing point cloud data from different time periods, it is possible to track changes in bridge parameters, providing insights into its health and facilitating effective lifecycle management.

Data availability

The dataset used in this study has been published on Mendeley Data for public release (Fang et al., 2024, https://doi.org/10.17632/znxxsgn2ky.1).

CRediT authorship contribution statement

Yunping Fang: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Stergios-Aristoteles Mitoulis:** Writing – review & editing, Supervision. **Daniel Boddice:** Writing – review & editing, Data curation. **Jialiang Yu:** Writing – review & editing, Methodology. **Jelena Ninic:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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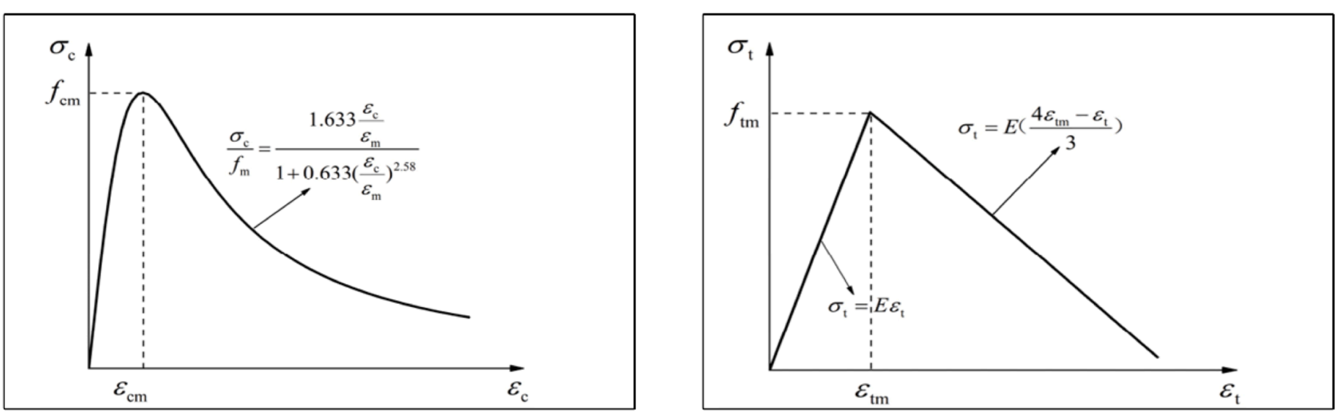
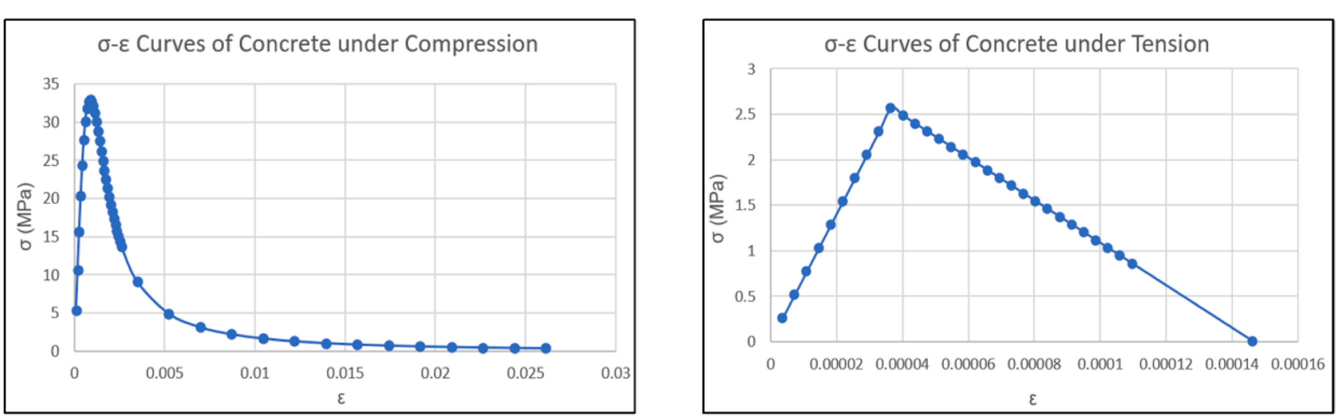
Appendix 1: σ-ε Curves of Concrete.

Fig. A1



(a) σ-ε curve of concrete under compression (left) and tension (right) (Theoretical)



(b) σ-ε curve of concrete under compression (left) and tension (right) (in ABAQUS)

Fig. A1. σ-ε curve of concrete [69].

The stress-strain curve of the concrete (see Fig. A1) is used for the input to ABAQUS.

Appendix 2: σ-ε curves of ordinary strength steel rebars.

Fig. A2

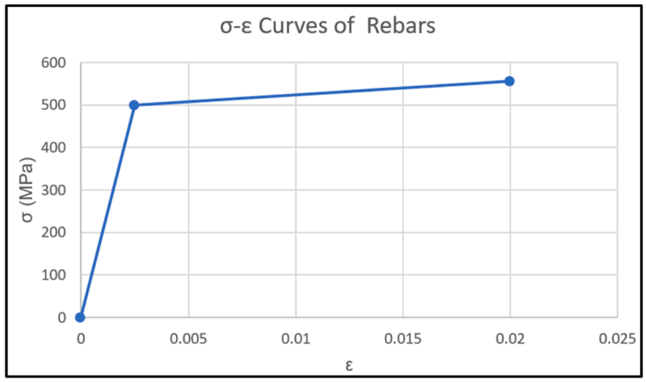


Fig. A2. σ-ε curve of Rebars.

The stress-strain curve of the rebars (see Fig. A2) is used for the input to ABAQUS.

Appendix 3: The load determination process.

Pedestrian Load on Walkways q_p: According to the Eurocode 1 [77], the uniform load for pedestrian areas (such as walkways on bridges) is 5 kPa.
Water Load in the Canal q_w: The uniform load exerted by the water in the canal depends on its depth and the density of water. Given:
Water density (ρ) = 1000 kg/m^3
Gravitational acceleration (g) = 9.81 m/s^2
Water depth (h) = 1.6 m
The water load is calculated as:

$$q_w = \rho g h = 1000 \times 9.81 \times 1.6 = 15.7\ kPa$$

Therefore, the uniform load from the water is 15.7 kPa.
Boat Load in the Canal q_b: Assuming the deck boat are approximately 5 ms in length and 2 ms in width with an estimated total weight of 2000 kg (including passengers and furnishings), the load distribution over the boat's contact area with the water is calculated as follows:
Total weight = 2000 kg
Contact width = 2 m
Contact length = 5 m
The boat load t is:

$$q_b = \frac{total\ weight \times g}{contact\ width \times contact\ length} = \frac{2000 \times 9.81}{2 \times 5} = 1.96\ kPa$$

Thus, the uniform load exerted by the boat on the water is approximately 1.96 kPa.

Data availability

The data is made openly avaible at https://doi.org/10.17632/znxxsgn2ky.1 which has been stated in the manuscript.

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